

CE 3890 – Report

INTRODUCTION:

This report summarizes and highlights the challenges and experiences faced throughout the class CE 3890 – Special Topics. This report includes the discussion and the process of building a segment of an aqueduct meant to imitate the famous Roman Arches in Segovia, and the creation of a water feature that was heavily influenced by Gaudi and his architectural designs. The report will dive into the construction of the Roman aqueduct segment, then the water feature. Then, it will go into the discussion of the Sagrada Familia and its connection to the course title, and a reflection of the trip and how it ties back to the course and understanding of civil engineering and other personal reflections.

Additionally, this report will include sketches showing the different periods of construction of the Montserrat site, and the Basilica construction. Followed by discussion of the different periods of construction in reference to the sketches.

SECTION 1: AQUEDUCT AND WATER FEATURE

Aqueduct design

Beginning the course with a project like the aqueduct in a group that was not entirely sure what the plan was going to be, introduced the idea for creativity. The first design that came to mind included 3d printing the entire arch and the column portions of the aqueduct. Our group found this to be the most practical way to go about the assignment, as we figured multiple iterations of the structure would need to be constructed and changing the dimensions in our columns or blocks through SolidWorks would be the easiest to go about it. We planned and discussed the best way to delegate the work, which splitting the work into 3 portions: the CAD designing portion of the voussoirs and columns, the printing of the designs, and the hands-on portion in lab with the scaffolding of the aqueduct arch.

When designing the CAD models for the arches, our team had hoped to find online models of any columns that we liked and printing them out, This would let the CAD portion be pretty simple and not hard to do, since the only real CAD portion would be designing the wedge blocks with the proper dimensions. However, since our dimensions did change a decent amount our team decided that although it was more work, CAD-ing the entire design would be the most optimal.

The CAD software that was used for the columns and arches was SolidWorks, which is great for sketching out unique (nonorganic) structures and extruding them to the proper dimensions. In the figure below shows the two models that were first designed, our column and our arches in one singular assembly.

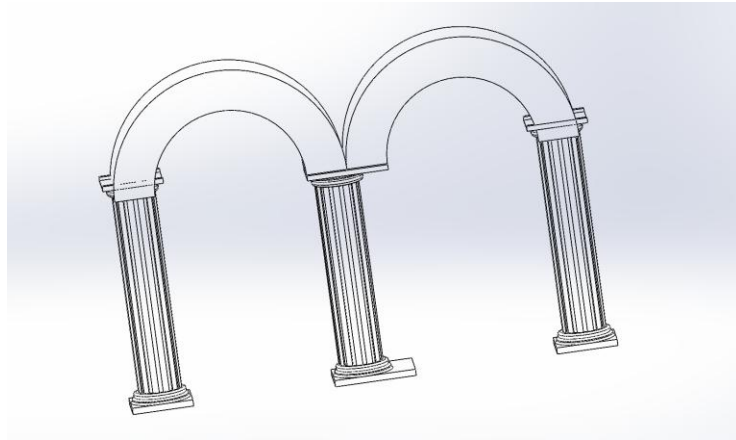


Figure 1. Assembly of initial Aqueduct Arches and Columns on SolidWorks

It is important to note that a challenge that our team faced during this project was agreeing on the right dimensions. I think our team had some communication issues, which led to designing with random dimensions and hoping they worked. We initially planned for around 10" columns and planned to have 6.5" arches. This would account for around 16.5" of our aqueduct height and then we could have room for our concrete to reach the proper height.

Our team had disagreed on the arches thickness and the arch height, which led to a change of dimensions entirely for both the columns and the arches. We finalized our height for our arches to be

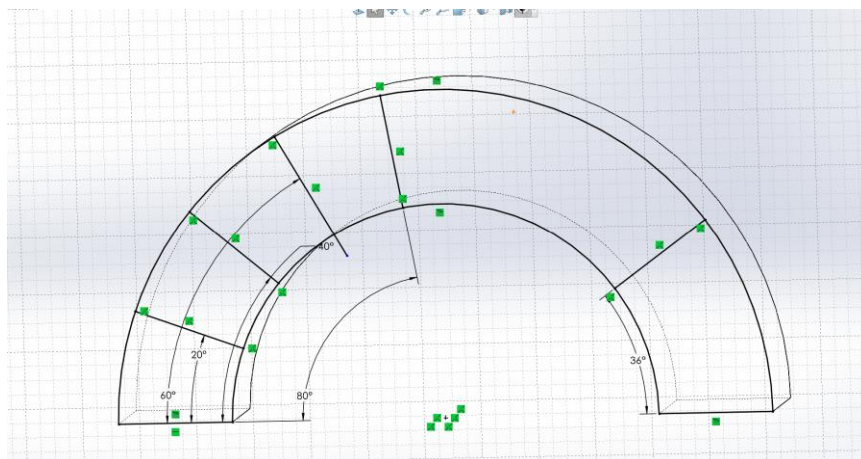


Figure 2. Arch design showing each wedge block is at 20 degrees of a semi-circle

In terms of the construction of the arches, we designed two arches to span the length of 2 feet, and each arch would have the same height and the same amount of voussoirs. We ended up with 9 stones for each arch as a design choice, so that we can showcase our arches are fully reliant on the compression to keep it together. The simplest way to design these arches was to sketch out a semicircle with an outer and inner radius that we wanted to be our height (outer radius) and inner radius would be the difference between our height and our thickness (3"). Below shows two figures, which were the final CAD designs for our wedge block models. It is important to note that for printing purposes we spaced each block equally, so each block was 20 degrees apart which leaves us with 9 identical wedge blocks for our design ($180/20 = 9$). This makes it significantly easier to print our blocks as we can 3D print the same block as many times as necessary and in the case of a failed print, we would not need to deal with the hassle of identifying which blocks we need to print.

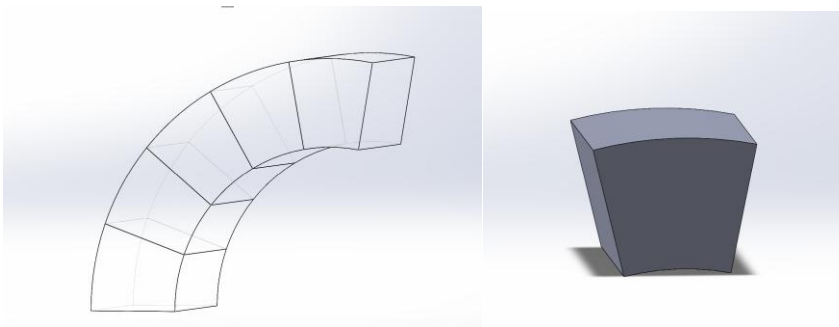


Figure 3 and Figure 4. Sketch showing half of the voussoirs together, and Figure 3 shows the CAD model of one wedge block

Switching over to the column design, after changing the dimensions of our arches, a redesign of our columns was necessary. To reiterate, our initial plan was to find columns online, but since models online are difficult to change the dimensions of the shaft alone, we CADded our own doric column. Since our plan included 2 arches, we would need to have 3 separate columns, with one holding up both arches. This idea of holding up the arches gave us the idea to print a column that included a man holding up the arches. Our column designs were already Greek-inspired as the columns are doric (Greek columns), so the man holding up the arches felt on theme for our project. It is meant to be similar to Atlas holding up the sky (Greek mythology). The following three figures shows the column we designed, the imported atlas statue, and the initial column we wished to parameterize initially(Corinthian column).



Figure 5 - Figure 7. Column designs that were used throughout the column printing process. Figure 6. was scrapped. Figure 5 is the final version of the self-designed doric column

Moving on to the printing aspect of the columns, when our team decided to change the arch height, we decreased our column height, which made it easier to print on campus as heights bigger than 9” are too large for some printers. After changing our heights to about 8-9” for each column we printed out the side columns first and were left with a really block-y design, which led to a redesign of our column and left us with our final design. The following figures show our failed prints, where 1) our doric column was not as emphasized and seemed too rectangular 2) our Atlas statue was too small and 3) our old column was too large after redesign and shows signs of a poor print job.

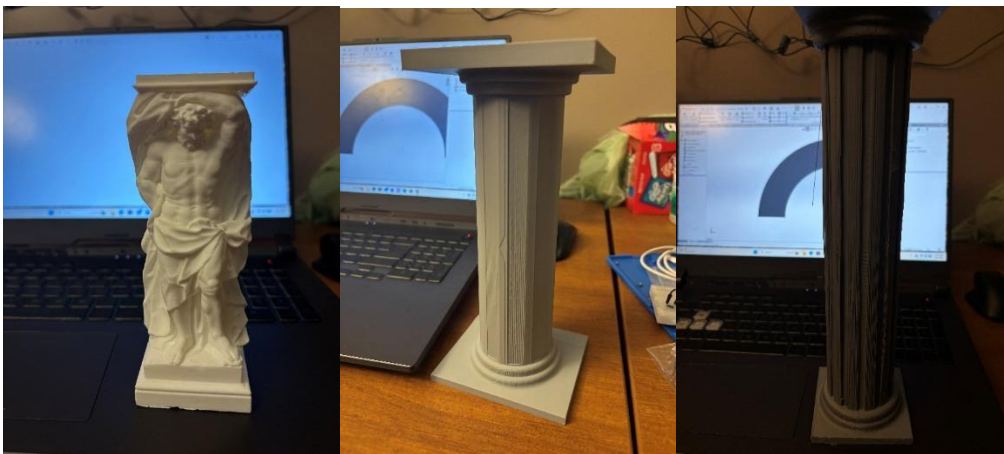


Figure 8-Figure 10. The 3d printed column with our old iteration, showing how some prints were more poor than others

Overall, the printing job was fairly tedious, and the CAD portion was also time consuming.

The scaffolding of our aqueduct was also another challenge for our team. Our team was unable to complete the scaffolding in time due to a prolonged time in deciding how to make our scaffolding. Our initial plan was to cut small pieces of balsa sticks and glue them

together as they were bent. However, our sticks endured too much stress while bent and kept breaking. It took a while for our team to finally jump ship and use Styrofoam for our scaffolding. This made it easier for us to just put our blocks together and keep them up before pouring concrete. Once we poured concrete and it cured, we were able to remove our scaffolding which included wood surrounding the arches, our arch portion was complete, and we were ready for test day. We did have 3 wedge blocks separated from our concrete arch which made the aqueduct questionable, but after constructing it together the arch still held up.

On Test Day, our arches were great and could have water pour down, and the height was correct as well. All ran smoothly for Aqueduct Group 2.



Figure 11 and Figure 12. The final aqueduct for our group, which shows the fully cured concrete and the arches.

Water Feature

Moving on to the design and construction process of our water feature, group 5 of the water features decided to go with an explicit Gaudi-inspired fountain. We found a lot of interest in the lizard from Parc Guell. Our initial planning process was rough, as we were not decisive in a height nor what materials we wanted to include. The consensus before the trip was to CAD the lizard to the best of our ability and cover it in clay. Since we wanted a fountain, our team was ambitious and planned on connecting a tube in the inside of the lizard to help stream water out of the lizard. Additionally, we wanted to improvise more from the lizard and emphasize the tail of the lizard, which meant having a tail wrap up to a basin

that would be higher than the lizard, which would then give it more of a dragon look. We also wanted to add wings for the lizard to transform it into a dragon.

Coming back from the trip, we had to rethink our entire plan. None of us had the experience working with a pump to power the fountain for 5 gallons of water, so we decided to just let gravity do the work and increase the initial height where water would flow. Additionally, initial plans to self-design the lizard were thrown out the window, due to Gaudi's lizard having lots of organic features. Our team was aware of this when starting the project, but we assumed we would have more time to use another 3D software to make the lizard called Blender. However, finding a model online was sufficient enough for our group as we could add more elements to cover for the height that we would miss out from self-designing the lizard. In comparison to other groups, I believe my lack of art skills set our group behind by a lot, as we only had a printed lizard until the week before our showcase.

Printing the lizard was more straightforward than the columns, as we did not have required heights to account for. Additionally, the clay allowed us to be flexible in terms of constructing our piece as we can have our build more inclined to be taller. We had a failed print, but we simply used the dimensions from that failed print to think of ideas of what we could do with the lizard once it was made.

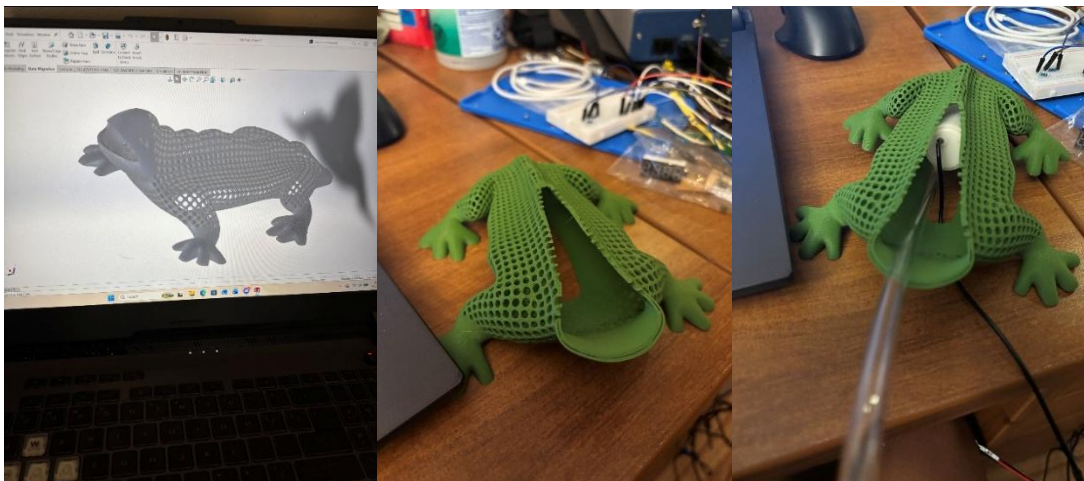


Figure 13- Figure 15 Cad of Lizard and failed print with small pump, showing initial plans when building water feature

Reiterating that the clay really allowed our group to be flexible with the way we went about building our lizard. We could add random stuff onto the build to imitate Gaudi and the clay stuck to it well.

Moving onto inspirations for this construction, once our team had finally printed our lizard and we found time to come into the lab to work on our build, our team decided to use rocks into the clay of the lizards body and later onto the rest of the clay portions to imitate the bumps on the lizard in parc Guell. We also put an excessive amount in comparison to

the real thing as a way to compensate for the lack of color in our build. The clay, however, was a perfect color for Casa Mia in Barcelona. Our group used an Aquafina bottle to have the lizard stand over that would be covered in clay. The clay structure with the water can, and also with the balsa stick (the sharp clay column on the far left) were inspired by the Casa Mia in Barcelona, as we found it interesting that the city tried to destroy his organic art because it was bulging out into the sidewalk.



Figure 16 & Figure 17. Gaudi lizard and our lizard side to side, showing how similar the two structures are.

Our biggest challenge in this portion of the construction of our water feature was time. By the time our team had found the time to put clay and make the water feature resemble its final look, we were very close to Test Day. This gave us the fear that our clay would not dry in time, and so our team improvised more towards the end of constructing the water feature.

To start off, we added broken seashells, broken pieces of plastic (red and blue shards on the top of the lizard) to resemble Gaudi's use of Trencadis, which was talked about in the Fab Lab in Barcelona. Trencadis is a form that Gaudi uses in his art a lot where he takes broken fragments and orientates them intricately to make a put together a specific shape. Figure 18 shows an example of broken fragments being aligned together.



Figure 18. Trencadis example, with blue and green shards

The final finishing portion of the water feature before testing includes making a shiny finish with epoxy. We were not sure if our water feature looked put together, so we had thought of other methods for making the clay look better, which led us to epoxy. This was, in our opinion, a great method because it closed any leaks that the clay had from cracking for the most part, as well as giving the clay a shiny look. Once the clay and epoxy set in and cured, we had left over epoxy, which led us to add seashells around our built. Once we had finished our water feature, we were ready for testing with our water feature being around 12". Figure 20 and 19 show the water feature before and after the epoxy, which shows the clay looking similar to clay after its been kilned, which looks more finished than other clay water features.



Figure 19 and Figure 20. water features after construction

On Test Day, our water feature worked nicely, picking up the water from the aqueduct and into the basins which poured out to the mouth of the lizard into the final bowl. In terms of improvements in our construction, we could have made our bowls a bit larger so it could hold more water, but the water feature held up well overall.

SECTION 2: THE SAGRADA FAMILIA

The construction that I was astonished by during the trip was the Sagrada Familia in Barcelona. It was a huge cathedral that was amazing in person, and I believe it to be the showcase of why our trip connects to our course. The Sagrada Familia is one of the biggest feats in civil engineering, especially since it was Gaudi's last huge project that he never got to finish. So many of his designs that we saw inside seemed so intentional and it was unbelievable to see in person. The Sagrada Familia itself especially highlights the use of engineering principles, in particular the ropes with loads that show how the Sagrada Familia should be oriented.

Additionally the Sagrada Familia highlights the historical side that comes with civil engineering. With Gaudi starting the construction in 1882, it shows how more and more of engineering improvements within the world can be used to build this construction that has been planned over 150 years ago. The biggest surprise in terms of history with the Sagrada Familia is the fact that the cables that Gaudi used to plan the construction of the towers was very good and shows how amazing engineers back then thought to construct buildings too.

Moving onto other portions of the Sagrada Familia, the structural parts within the cathedral also have their parts that they play. Gaudi used catenary arches to transfer loads, which allows for the cathedrals' organic structure on the outside, without using other systems that cathedrals back then had, like flying buttresses (like in Notre dame).

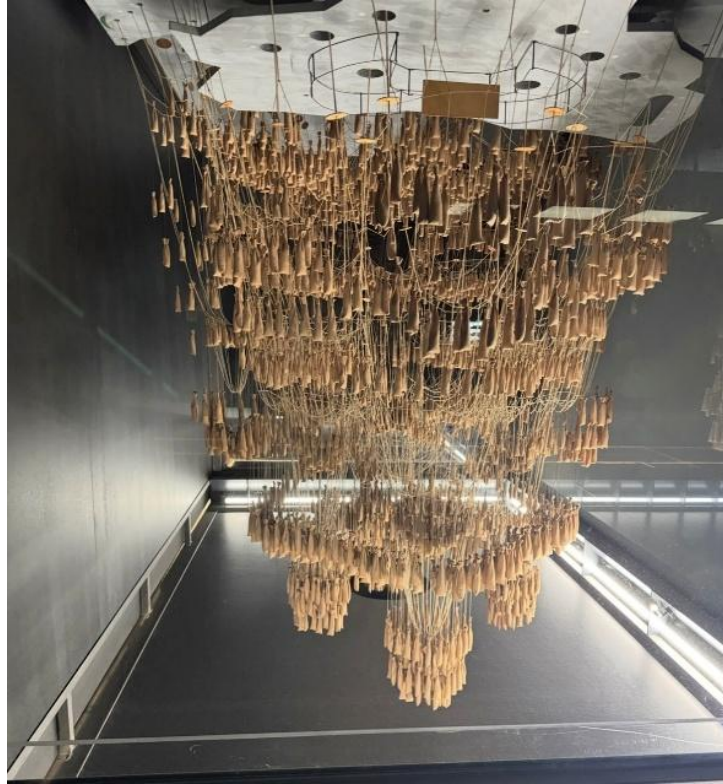


Figure 21. The basement of the Sagrada Família, displaying Gaudi's cable and distributed weights to figure out the composition for the towers

Looking inside the cathedral again, the pillars along the interior are very organic looking, and according to the tour guide were used and made to resemble trees. Having a large trunk that then splits off around the top, as seen in the figure, is an amazing feature of the Sagrada Família as it allows for more forces to be distributed from the top to the bottom. So many of these tree columns were filled inside the cathedral, it is basically a forest like structure with the number of “trees” in the interior. Furthermore, since the Sagrada Família follows the basic shape of most cathedrals, where the outline is a Cross, the cathedral has different points that have a lot more loads to bear. This makes the tree-like columns have to take on heavier or lighter loads. Gaudi worked around this by making the columns with different material, so that the columns had different material as well. All of these stability features are so innovative and so well designed by Gaudi, it shows how an engineer can really think about every singular portion of their design and make it their own unique style and still complete what needs to be done (have a structurally sound cathedral).



Figure 22. Sagrada Familia's Tree like columns, with the top extending out to the top of the building

In terms of the artistic features of the Sagrada Família, this also highlights a portion of the class like the history and innovation in engineering. Although you can say Gaudi using cables was innovative in terms of engineering, Gaudi also used his brain to create intricate styles inside the church that are unforgettable. The first thing that comes to mind is the stained glass of the cathedral, and the colors that Gaudi used. To start off Gaudi, intentionally used the position of the Sagrada Família to his advantage, as he used the sunlight to create colors from the stained glass that made Sagrada Família a unique experience. To explain this feature in more detail, Gaudi had made one side of the cathedral have warmer colored stained glass that would light up the cathedral during a certain time of the day. However, during the other portion of the day, the sun illuminated the other side, which had cooler colors for the stained glass, which contrasted with the other side. This was amazing to see, and I believe it to be an excellent example of an innovative idea in engineering.

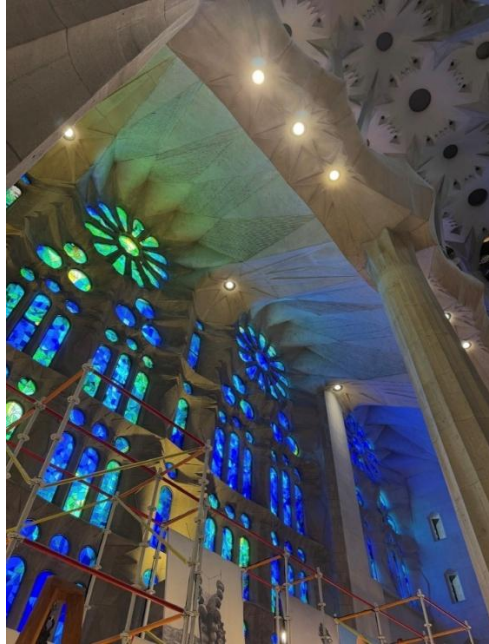


Figure 23. Sagrada Familia stained windows with cooled colors shining out

Lastly, talking about religion, this highlights the history that comes with engineering again. Gaudi built this entire building, which was used for religious purposes. This tells us a lot of how religion and history are embedded in our designing portion of engineering. In the Sagrada Família alone, Gaudi included lots of religious figures and had statues for them and names listed inside and outside of the Sagrada Família. This highlights the work that Gaudi did, especially with the part where the entire Sagrada Família is cross-shaped for religious purposes.



Figure 24 and Figure 25. The Sagrada Família from the outside showing the organic shapes

SKETCHES AND MONSERRAT:

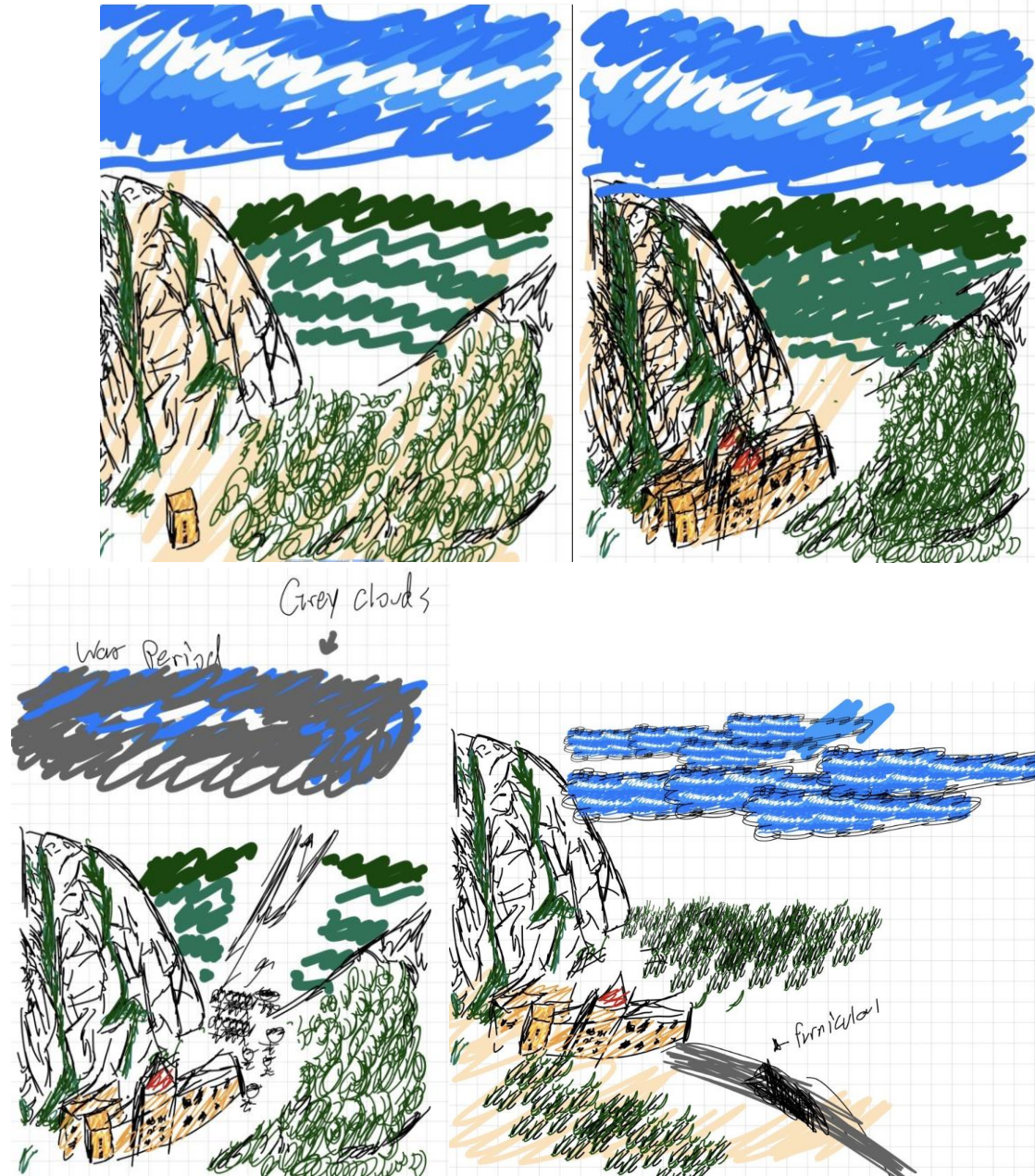


Figure 26 - Figure 29. 4 periods of construction in Montserrat. Early period with just monk settling in, then then construction of the basilica and the monastery, then the war period, and finally the modern period

Although Monserrat was a city that I missed out on, the construction that led to what Monserrat is something to still learn. The development of Montserrat came to be from different periods of construction, where Monserrat drastically changed. These periods were defined by events that were happening during the time, which is why Monserrat connects nicely to our class.

Montserrat lays on a mountain, which made it difficult for communities to grow a town there without carefully planning for resources. However, the earliest dates and period when people moved onto the mountain were very early on (during the year 1025). During this period of construction, a person named Abbot Oliba had founded a monastery, which influenced Monserrat's construction, as it began as a religious site in this area (Catalonia). This led to the inhabitants of Montserrat to consist primarily of monks that were up in the mountains for the monastery.

The next period of construction that was a defining moment for Monserrat included the construction of the basilica, which began in the 15th century. This again highlights how much influence religion had on Montserrat, and to know that during this period, both the notable basilica and the monastery were being constructed, which are central buildings/structures in Montserrat at the time.

During the years of war (1808), the monks that were in Montserrat fled, and took shelter in other cities like Mallorca. This allowed for the armies to occupy Montserrat, take shelter there and made Montserrat's monastery a military building from this period, where it was a location for weapons to be sent around the Catalonia area. The basilica and monastery had to be fixed later, a couple years later (18040-1850s) as war had led to the building's destruction, but the rebuilt buildings led to the decision to preserve its original design and keeping it as close as its original design.

After this period of war was also significant because of the rebuilt Monastery and the end of the war with the French, the monks that had fled and the Castilian community that were in Montserrat were no longer there, so around 1840's, by this point Montserrat had become more of an independent place.

With the slow growth of Montserrat, the modern construction of Montserrat came with the first funicular de Sant Joan was constructed in 1918. This led to the city of Montserrat adopting more modern city structures.

SECTION 3: REFLECTIONS

The class affected my understanding of civil engineering vastly. I had never taken into account how much thinking can be put into projects like city buildings or buildings in general. Perhaps it was only looking at Gaudi's structures, who definitely put a lot of effort

into all of his projects, but the limit at which a person can design a building was never something I thought could be so high. Especially when looking at the Sagrada Familia, a building whose construction took years and generations, the building has so many intentional features that keep adding more to the building itself. Additionally, I think the class helped me understand how much constraints or limitations go into making something. For example, I feel like when constructing the aqueduct, there were so many things our group was attentive to, but even with that, in the end we forgot to account for the concrete expanding until the day we poured the concrete, which led to last minute fixes, which affected our project in the end.

I have so much more appreciation for architects, specifically Gaudi. I think this class focused a lot on him and I am glad it did, because he truly does show me and most definitely others just how much effort one can pour into their project.

The trip has allowed me to focus a lot more on decision making in general. I did not really think about myself as impulsive or not the best decision maker, but after this trip, I definitely think there is room for improvement in both planning and decision making. I think I try not to get people to worry about me and be able to take care of myself, but in a trip like this, it reminds me it is okay to reach out and ask someone for help.

CONCLUSION:

Overall the class focused on Gaudi architecture, which was a different viewpoint that I had not thought about, in particular the architectural stance when it comes to designing buildings, and I think during the construction and planning period of the aqueduct and water feature, I wanted to try and challenge myself, which is why I think I spent a lot of effort in both projects and did not mind putting in the amount of work I did in both projects (I enjoyed making the water feature more than the aqueduct but that was because 3d printing at fgh was tedious).